

Assignment – Docker Class4

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root@ip-172-31-68-87: ~
root@ip-172-31-68-87:~# mkdir new-app-dir
root@ip-172-31-68-87:~# docker run --name v2 -v /home/ubuntu/new-app-dir:/opt/new-app-dir amaz
onlinux:latest
root@ip-172-31-68-87:~# docker run --name v3 -v /home/ubuntu/new-app-dir:/opt/n
ew-app-dir -it amazonlinux:latest
bash-5.2# ls
bin    dev    home  lib64  media  opt    root  sbin  sys  usr
boot  etc    lib   local  mnt    proc   run   srv   tmp  var
bash-5.2# cd opt/
bash-5.2# ls
new-app-dir
bash-5.2# cd new-app-dir/
bash-5.2# touch f1 f2 f3 f4 f5
bash-5.2# ls
f1 f2 f3 f4 f5
bash-5.2# root@ip-172-31-68-87:~# pwd
/root
root@ip-172-31-68-87:~# cd /home/ubuntu/
root@ip-172-31-68-87:/home/ubuntu# ls
new-app-dir
root@ip-172-31-68-87:/home/ubuntu# cd new-app-dir/
root@ip-172-31-68-87:/home/ubuntu/new-app-dir# ls
f1 f2 f3 f4 f5
root@ip-172-31-68-87:/home/ubuntu/new-app-dir# #Now even if container v3 gets d
eleted data will be preserved in dockerhos
root@ip-172-31-68-87:/home/ubuntu/new-app-dir#

```

```

root@ip-172-31-68-87: ~
amazonlinux:latest c4543691c8ce 227MB 59.3MB U
root@ip-172-31-68-87:~# vi Dockerfile
root@ip-172-31-68-87:~# cat Dockerfile
FROM amazonlinux:2
MAINTAINER CS

root@ip-172-31-68-87:~#
root@ip-172-31-68-87:~#
root@ip-172-31-68-87:~# docker build -t amazonlinux:2 .
[+] Building 0.4s (5/5) FINISHED docker:default
=> [internal] load build definition from Dockerfile 0.0s
=> => transferring dockerfile: 72B 0.0s
=> WARN: MaintainerDeprecated: Maintainer instruction is deprecate 0.0s
=> [internal] load metadata for docker.io/library/amazonlinux:2 0.1s
=> [internal] load .dockerignore 0.0s
=> => transferring context: 2B 0.0s
=> CACHED [1/1] FROM docker.io/library/amazonlinux:2@sha256:dc69cc 0.0s
=> => resolve docker.io/library/amazonlinux:2@sha256:dc69cc77521b5 0.0s
=> exporting to image 0.1s
=> => exporting layers 0.0s
=> => exporting manifest sha256:146d010be8e60ca14ad7a43991e2ceeb81 0.0s
=> => exporting config sha256:3318d3e3fc4c66d63f895d5594cfa2fa5078 0.0s
=> => exporting attestation manifest sha256:284d8d4dd3c66d2933a0d7 0.0s
=> => exporting manifest list sha256:313e6118130839d4c914a33cb92f1 0.0s
=> => naming to docker.io/library/amazonlinux:2 0.0s
=> => unpacking to docker.io/library/amazonlinux:2 0.0s

1 warning found (use docker --debug to expand):
- MaintainerDeprecated: Maintainer instruction is deprecated in favor of using label (line 2)
root@ip-172-31-68-87:~# docker images

```

IMAGE	ID	DISK USAGE	CONTENT SIZE	Info →	U In Use	EXTRA
amazonlinux:2	313e61181308	251MB	63MB			
amazonlinux:latest	c4543691c8ce	227MB	59.3MB		U	

```

root@ip-172-31-68-87:~# docker run --name AL1 -it amazonlinux:2
bash-4.2# root@ip-172-31-68-87:~#
root@ip-172-31-68-87:~# docker ps

```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
e14759355ea3	amazonlinux:2	"/bin/bash"	About a minute ago	Up About a minute		AL1
8588f2dd0fea	amazonlinux:latest	"/bin/bash"	38 minutes ago	Up 38 minutes		v3
101726b61773	amazonlinux	"/bin/bash"	44 minutes ago	Up 44 minutes		newone

```

root@ip-172-31-68-87:~#

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root@ip-172-31-68-87: ~
root@ip-172-31-68-87:~# docker build -t amazonlinux:26 .
[+] Building 40.3s (6/6) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 93B
=> WARN: MaintainerDeprecated: Maintainer instruction is deprecated in favor of using label (line 2)
=> [internal] load metadata for docker.io/library/amazonlinux:latest
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/2] FROM docker.io/library/amazonlinux:latest@sha256:c4543691c8ce4157cbfe7d10170106eccb9dedd61f022980f062788d00e8af51
=> => resolve docker.io/library/amazonlinux:latest@sha256:c4543691c8ce4157cbfe7d10170106eccb9dedd61f022980f062788d00e8af51
=> => sha256:a2488c40110fb2b90385f44d9af6173894e14350935e38653aee164551cb70d6 54.57MB / 54.57MB
=> => extracting sha256:a2488c40110fb2b90385f44d9af6173894e14350935e38653aee164551cb70d6
=> [2/2] RUN yum install git -y
=> exporting to image
=> => exporting layers
=> => exporting manifest sha256:91ed29062aae3d5c6b62daca4502a0383327c44ec0d89b4b793c566e9402750c
=> => exporting config sha256:b00e5a00041215d095b5cc199d47d61e251ba308d902aa465cda8c123d9ae3e8
=> => exporting attestation manifest sha256:8a8d4faf0af3924b056f89830a671107308d7c0e9d6964fe26670617c917ef29
=> => exporting manifest list sha256:eb934a22db29ac58190e353e9654dbf1ba222d60fe43db7b8db4c3c3cf7db936
=> => naming to docker.io/library/amazonlinux:26
=> => unpacking to docker.io/library/amazonlinux:26

1 warning found (use docker --debug to expand):
- MaintainerDeprecated: Maintainer instruction is deprecated in favor of using label (line 2)
root@ip-172-31-68-87:~# docker images

IMAGE          ID                DISK USAGE  CONTENT SIZE  EXTRA
amazonlinux:26 eb934a22db29      626MB       199MB
root@ip-172-31-68-87:~# docker run --name v1 -it amazonlinux:26
bash-5.2# git --version
git version 2.50.1
bash-5.2# root@ip-172-31-68-87:~# docker ps
CONTAINER ID   IMAGE                COMMAND                  CREATED        STATUS        PORTS        NAMES
dd84398d8830   amazonlinux:26      "/bin/bash"             34 seconds ago Up 34 seconds        v1
root@ip-172-31-68-87:~# cat Dockerfile
FROM amazonlinux
MAINTAINER CS
RUN yum install git -y

```

```

root@ip-172-31-68-87: ~
root@ip-172-31-68-87:~# vi Dockerfile
root@ip-172-31-68-87:~# cat Dockerfile
FROM debian:trixie
MAINTAINER CS
RUN apt update
RUN apt install git -y

root@ip-172-31-68-87:~# docker build -t debian:trixie .
[+] Building 22.8s (7/7) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 111B
=> WARN: MaintainerDeprecated: Maintainer instruction is deprecated in favor of using label (line 2)
=> [internal] load metadata for docker.io/library/debian:trixie
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/3] FROM docker.io/library/debian:trixie@sha256:3352c2e13876c8a5c5873ef20870e1939e73cb9a3c1aeba5e3e72172a85ce9ed
=> => resolve docker.io/library/debian:trixie@sha256:3352c2e13876c8a5c5873ef20870e1939e73cb9a3c1aeba5e3e72172a85ce9ed
=> CACHED [2/3] RUN apt update
=> [3/3] RUN apt install git -y
=> exporting to image
=> => exporting layers
=> => exporting manifest sha256:91cda07392f20533552301024f1a60beea228452fd2af19eb204744a7316242e
=> => exporting config sha256:1da61c8edc9693a64789866ecf7f636803d2a5f6f6ab286f974875b5e90c803f
=> => exporting attestation manifest sha256:ef2d27989a99d2aab53159b5edd093c637eab09354a05600289b9aa9dd60823e
=> => exporting manifest list sha256:47a5f9c53ecb72fclde9f79ald018ea7b2cbebbe58d8956bd17ba7f66ba462870
=> => naming to docker.io/library/debian:trixie
=> => unpacking to docker.io/library/debian:trixie

1 warning found (use docker --debug to expand):
- MaintainerDeprecated: Maintainer instruction is deprecated in favor of using label (line 2)
root@ip-172-31-68-87:~# docker images

IMAGE          ID                DISK USAGE  CONTENT SIZE  EXTRA
amazonlinux:26 eb934a22db29      626MB       199MB
debian:trixie  47e5f9c53ecb      433MB       121MB
root@ip-172-31-68-87:~# docker run --name dbn-trx-1 -it debian:trixie
root@f432244909c7:/# git --version
git version 2.47.3
root@f432244909c7:/# root@ip-172-31-68-87:~# docker ps
CONTAINER ID   IMAGE                COMMAND                  CREATED        STATUS        PORTS        NAMES
f432244909c7   debian:trixie       "bash"                 37 seconds ago Up 36 seconds        dbn-trx-1
dd84398d8830   amazonlinux:26      "/bin/bash"             10 minutes ago Up 10 minutes        v1
root@ip-172-31-68-87:~#

```

```
root@ip-172-31-68-87:~# vi Dockerfile
root@ip-172-31-68-87:~# cat Dockerfile
FROM ubuntu
MAINTAINER xyz
CMD date

root@ip-172-31-68-87:~# docker build -t ubuntu:v1 .
[*] Building 2.1s (5/5) FINISHED
=> [internal] load build definition from Dockerfile
=> transferring dockerfile: 75B
=> WARN: MaintainerDeprecated: Maintainer instruction is deprecated in favor of using label (line 2)
=> [internal] load metadata for docker.io/library/ubuntu:latest
=> [internal] load .dockerignore
=> transferring context: 2B
=> [1/1] FROM docker.io/library/ubuntu:latest@sha256:84e77dee7d1bc93fb029a45e3c6cb9d8aa4831ccfcc7103d36e876938d28895b
=> resolve docker.io/library/ubuntu:latest@sha256:84e77dee7d1bc93fb029a45e3c6cb9d8aa4831ccfcc7103d36e876938d28895b
=> sha256:689b91d88a0f4086057ec826027b128902ecf2b516be510371c115bc55da19a6 29.73MB / 29.73MB
=> exporting to image
=> exporting layers
=> exporting manifest sha256:16b909b6d6837c25ce9c7a61993a7ebba228c50287a847e5ef02ea9bda58197d
=> exporting config sha256:2a08980120b2d151182f29d047d2eab1970508cdc88afd4940fbc934aa2d1cd
=> exporting attestation manifest sha256:87b111bb0cc499c676d9ac5e2a08a6619b0a62c5a0c12966ffba68abd06812b
=> exporting manifest list sha256:93d01e0f9aac1d7be47097008a74fald7c24612af25d4906c5e524e00de4f55
=> naming to docker.io/library/ubuntu:v1
=> unpacking to docker.io/library/ubuntu:v1

2 warnings found (use docker --debug to expand):
- MaintainerDeprecated: Maintainer instruction is deprecated in favor of using label (line 2)
- JSONArgsRecommended: JSON arguments recommended for CMD to prevent unintended behavior related to OS signals (line 3)
root@ip-172-31-68-87:~# docker images

IMAGE                                ID                                DISK USAGE  CONTENT SIZE  EXTRA
amazonlinux:26                      eb934a22db29                     626MB       199MB         U
debian:trix                         47e5f9c53ecb                     433MB       121MB         U
ubuntu:v1                           93d01e0f9aac                     117MB       29.7MB
root@ip-172-31-68-87:~# docker run --name UBC-1 -it ubuntu:v1
Fri Apr 10 19:35:46 UTC 2026
root@ip-172-31-68-87:~#
root@ip-172-31-68-87:~# docker ps -a
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS   NAMES
g057922f2567   ubuntu:v1  "/bin/sh -c date"       27 seconds ago  Exited (0)   26 seconds ago  UBC-1
f432244909c7   debian:trix  "bash"                 10 minutes ago  Up 10 minutes  dbn-trx-1
dd84398d8830   amazonlinux:26  "/bin/bash"            20 minutes ago  Up 20 minutes  v1
```

In this scenario, the `CMD` instruction is defined as `date`, which means when we execute `'docker run'`, Docker starts the container and runs `'date'` as the main process.

Since `date` is a short-lived command, it executes, prints the current timestamp, and exits immediately with a success status. Docker monitors this main process, and once it exits, Docker considers the container's execution complete and stops the container. So, the container stops because the process defined by `CMD` has finished execution, not because of any failure.

Q: Difference between AND operator & OR operator used in Linux

A: ‘And’ & ‘OR’ operators are binary operators used to control how commands execute based on the result of the previous command.

- AND: This binary operator returns true if both the operands are true, otherwise returns false.
- OR: This binary operator returns true if either of the operands is true or both the operands are true and returns false if none of them is false.